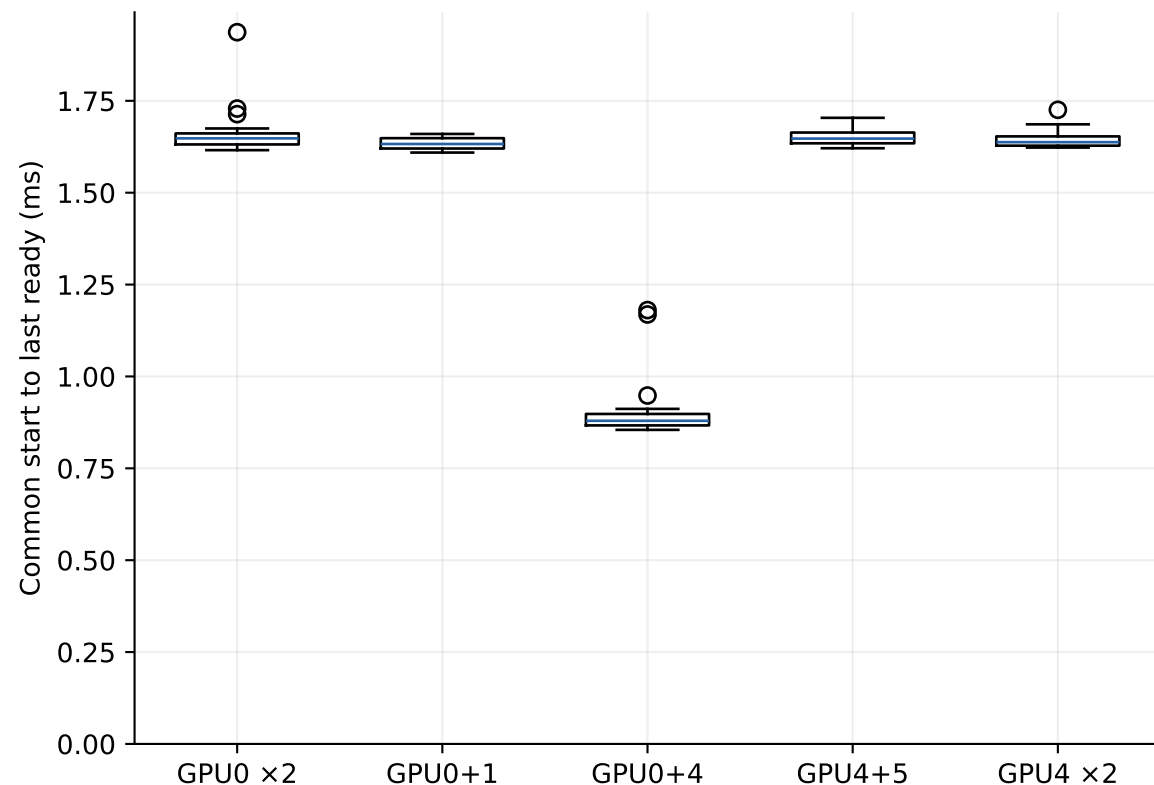
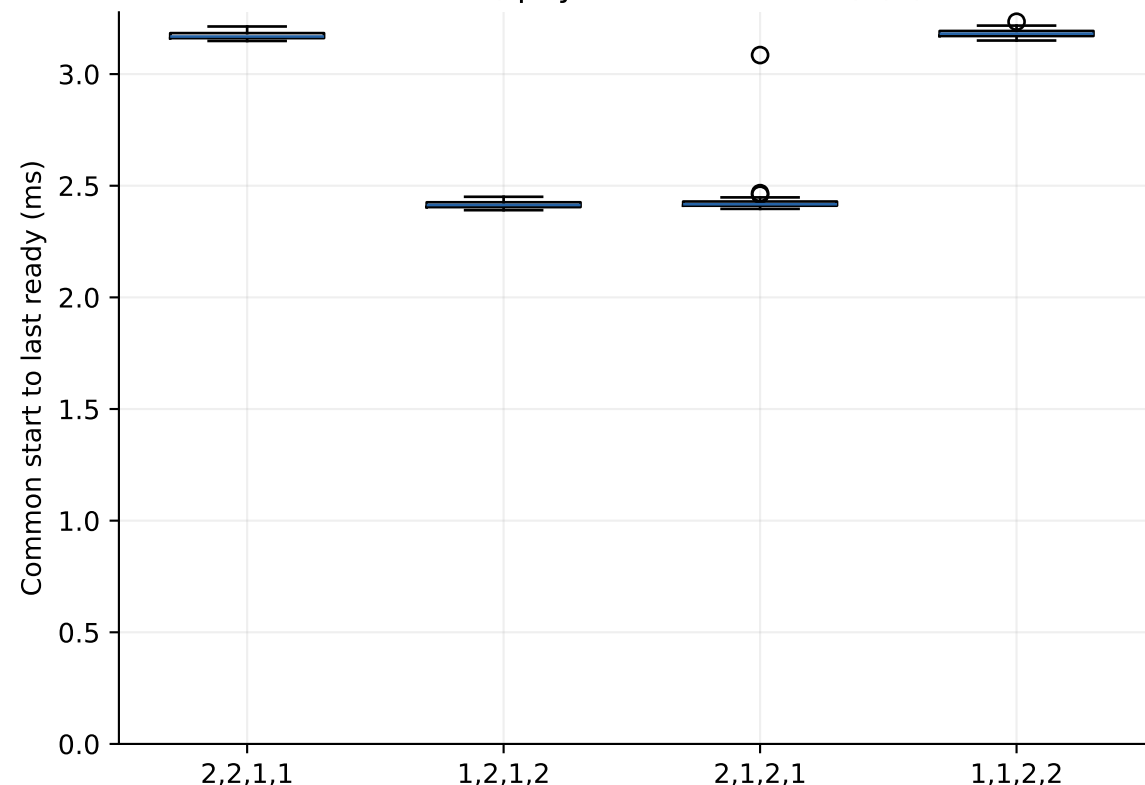


Isolated H2D, verified local NUMA pinned sources — 54 GiB pinned/node; spread addresses

Two 9-MiB fetches



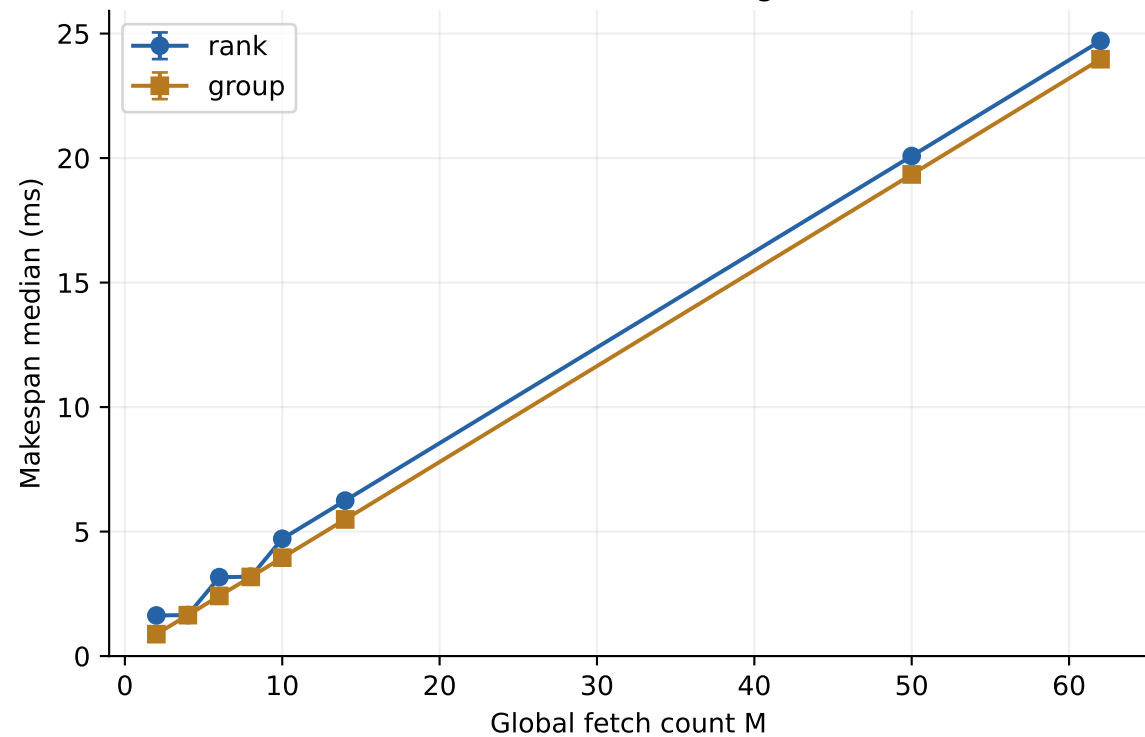
Six fetches; physical GPU order 0,1,4,5



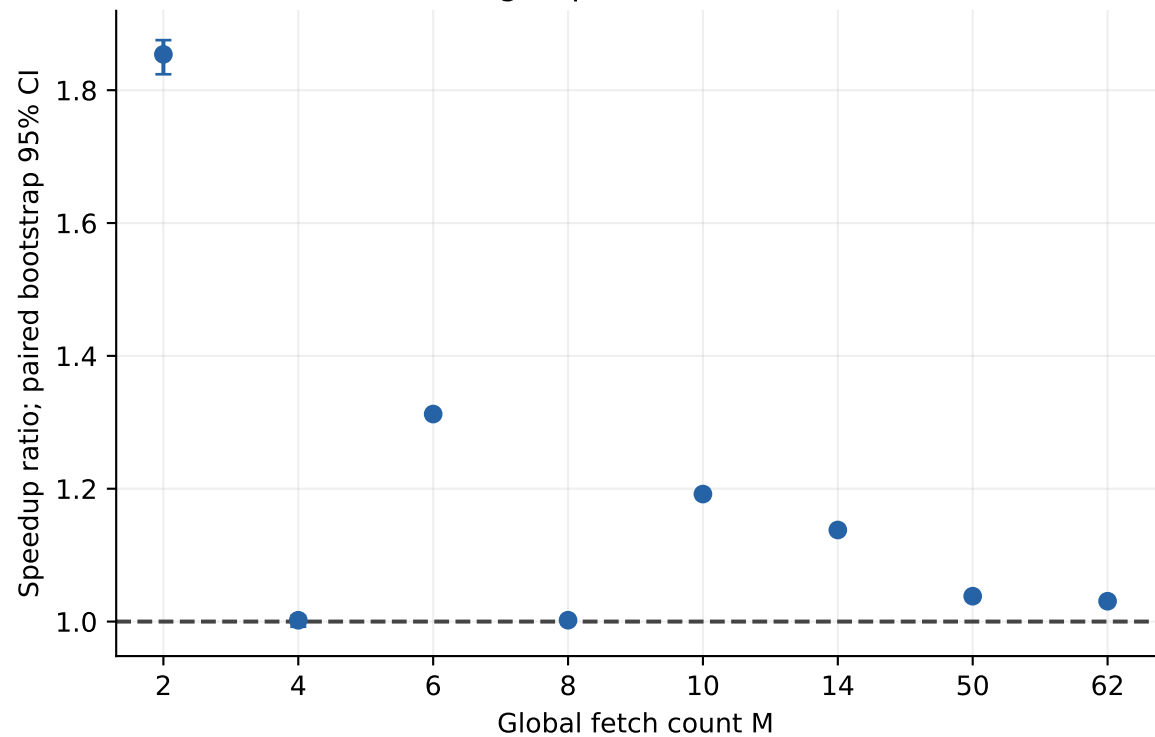
30 samples per cell after 5 warmups; randomized repeat blocks. Boxes: quartiles; lines: medians; dots: outliers.

One copy stream per GPU; M=4 and M=8 are negative controls
54 GiB pinned/node; spread addresses

Miss-count scaling

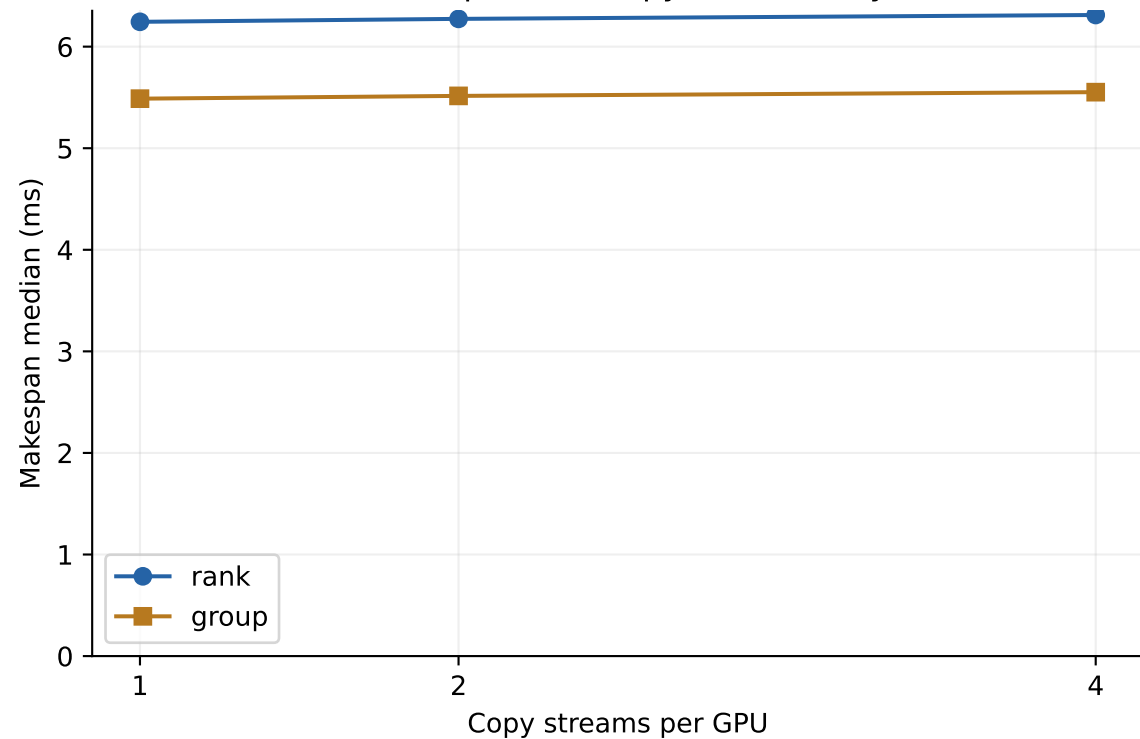


Rank/group ratio of medians



Remote pages verified; G2G overlap is a separate diagnostic
54 GiB pinned/node; spread addresses

M=14: per-GPU copy concurrency



M=6: source/overlap sensitivity

